NETMAC ANALYZER: MAC ADDRESS SNIFFER & PRIVACY CLASSIFIER

Comprehensive Final Technical Project Report

Course / Subject:	Computer Networks (CSEN3021)	**Team Number:**	Team 23 (Slot F1 + TF1)
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ABSTRACT

In modern local area networks (LANs) and wireless environments, physical station identification and link-layer frame routing operate at the Data Link Layer (Layer 2) of the OSI model using 48-bit Media Access Control (MAC) addresses. Traditional network monitoring and packet analysis software, such as Wireshark and tcpdump, monitor network adapters in broad promiscuous modes. When attached to standard physical network cards, these tools indiscriminately capture immense volumes of ambient broadcast, multicast, and background operating system telemetry (such as mDNS, SSDP, LLMNR, and NetBIOS). This ambient noise obscures actual client-to-gateway traffic flows and prevents administrators from establishing an idle baseline. Furthermore, standard diagnostic analyzers treat MAC addresses as passive hexadecimal identifiers without automatically interpreting their underlying administrative structure or privacy state, particularly with the widespread implementation of MAC Address Randomization in modern smartphone operating systems (such as Android 10+ and iOS 14+).

To resolve these challenges, this project presents **NetMAC Analyzer**, an automated, real-time Layer-2 and Layer-3 network packet capture and hardware address analysis framework developed in Python. The system binds directly to a local wireless hotspot virtual gateway interface (Microsoft Wi-Fi Direct Virtual Adapter #2 on subnet 192.168.137.0/24), isolating client traffic from external physical network noise. The framework enforces a deterministic zero-packet idle baseline upon startup, captures frames asynchronously, and executes a real-time bitwise decoding algorithm on the first octet of the MAC address to differentiate Universally Administered Addresses (UAA) from Locally Administered Addresses (LAA - Randomized Private MAC). It correlates link-layer physical identities with Layer-3 IPv4 endpoints and Layer-4 TCP/UDP transport flows, providing on-demand hierarchical protocol dissection and 16-byte aligned hexadecimal/ASCII memory dumps through a responsive web interface. Empirical validation across an isolated mobile hotspot session of 148 packets demonstrated 100% accuracy in detecting mobile MAC randomization, sub-15-microsecond parsing latency, and an automated test suite pass rate of 100%.

1. INTRODUCTION

1.1 Technical Background: The Open Systems Interconnection (OSI) model separates network communications into hierarchical functional layers. In this model, logical routing across networks occurs at Layer 3 (Network Layer) using Internet Protocol (IP) addresses, while physical station delivery across a local broadcast domain occurs strictly at Layer 2 (Data Link Layer) using Media Access Control (MAC) addresses. Standardized by the IEEE 802 committee, a MAC address consists of 48 bits (6 octets). Historically, these identifiers were burned into the physical read-only memory of network interface cards by manufacturers, forming globally unique Universally Administered Addresses (UAA). However, because mobile devices continuously transmit link-layer frames during network discovery and association, persistent hardware addresses allow commercial tracking systems and rogue stations to silently monitor users' physical locations. To preserve user privacy, modern mobile operating systems generate synthetic, temporary MAC addresses for Wi-Fi associations, designated under the IEEE 802 standard as Locally Administered Addresses (LAA).

1.2 Problem Statement: Traditional network monitoring software monitors entire physical adapters in broad promiscuous modes, capturing immense volumes of ambient broadcast and operating system telemetry. This ambient background chatter obscures actual

client-to-gateway communication flows and creates an information-overload barrier for network administrators conducting hardware-focused link-layer diagnostics. Furthermore, traditional analyzers treat MAC addresses as passive hexadecimal identifiers without automatically interpreting their underlying administrative structure or privacy state. There is an absence of specialized, lightweight analysis tools that can monitor an isolated network link (such as a mobile hotspot gateway), enforce a clean zero-packet initial state, filter out extraneous external noise, and provide automated link-layer privacy classification alongside Layer-3 IP flow correlation.

1.3 Objectives of the Project: The primary objectives of NetMAC Analyzer are:

- **Targeted Interface Binding:** Interface directly with kernel-level packet capture drivers (Npcap) to bind sniffing routines specifically to the virtual mobile hotspot interface (Microsoft Wi-Fi Direct Virtual Adapter #2), isolating client traffic from external physical Wi-Fi noise.
- **Zero-Packet Baseline Enforcement:** Enforce an idle baseline where the packet counter remains strictly at zero upon initialization and resets to zero upon clearing, capturing frames only when connected stations actively transmit or receive data.
- **Automated Bitwise Privacy Classification:** Extract 48-bit MAC addresses and inspect the first octet at the bit level (Bit 0 for Unicast/Multicast, Bit 1 for LAA/UAA) to automatically detect and flag randomized mobile device MAC addresses.
- **Hardware Provenance Resolution:** Map Organizationally Unique Identifier (OUI) prefixes against standardized manufacturer databases to identify physical network card vendors offline.
- **Multi-Layer Protocol Correlation:** Correlate Layer-2 MAC addresses with Layer-3 IPv4 addresses, Layer-4 transport protocols (TCP with state flags, UDP, ICMP), and Layer-7 application metadata (DNS queries and TLS Server Name Indication).
- **Transparent Low-Level Inspection:** Provide 16-byte offset hexadecimal and ASCII dumps for direct verification of frame bytes without memory truncation.
- **Interactive Web-Based Visualization:** Stream real-time frame summaries into an interactive web dashboard utilizing DOM virtualization to prevent browser memory exhaustion.

2. LITERATURE REVIEW

2.1 Evolution of Packet Capture Technologies: Packet sniffing originated in early shared coaxial Ethernet topologies where all connected stations physically received every transmission on the bus. In 1993, McCanne and Jacobson developed the BSD Packet Filter (BPF) [5], introducing an in-kernel register-based virtual machine that applies filtering predicates directly inside operating system kernel memory. This eliminated the need to copy unwanted packets into user space, drastically reducing CPU context-switching overhead and forming the foundation for libpcap and its Windows counterpart, Npcap [6].

2.2 IEEE 802 Addressing and Privacy Evolution: The IEEE 802-2014 standard defines the 48-bit Extended Unique Identifier (EUI-48) format [1]. The first 3 octets represent the Organizationally Unique Identifier (OUI) assigned to manufacturers, while the final 3 octets represent the device-specific identifier. Within the first octet ($Byte_0$), the IEEE embeds two control bits: Bit 0 (I/G Bit: 0 = Unicast, 1 = Multicast/Broadcast) and Bit 1 (U/L Bit: 0 = Universal UAA, 1 = Local LAA). Academic research by Martin et al. [3] and Vanhoef et al. [4] demonstrated that static factory addresses enable third-party profiling and tracking across wireless networks. In response, Android 10+ and iOS 14+ standardized connection-level MAC address randomization. By setting the U/L bit to 1 while keeping the I/G bit at 0, mobile operating systems generate non-trackable addresses that remain compliant with IEEE 802 standards.

Tool / Framework	Interface Isolation	Idle Startup Baseline	Bitwise MAC Privacy	User Interface
Wireshark 4.x [13]	Manual display filtering	Flooded with OS noise	Manual bitmask calculation	Heavy C++/Qt Desktop App
tcpdump CLI [14]	Promiscuous physical binding	Flooded with noise	Raw hexadecimal only	Terminal text output
Kismet Wireless [15]	Monitor mode raw RF	Passive radio logging	Probe request analysis	Heavyweight Web / CLI
NetMAC Analyzer (Ours)	Targeted Virtual Adapter	Strict Zero Baseline (0 Pkts)	Automated Real-Time LAA Flag	Lightweight DOM-Virtualized Web

Table 2.1: Comparative Summary of Packet Sniffing Tools.

3. INSTALLATION & REQUIREMENTS SETUP GUIDE

3.1 Setup Instructions for New Environments: Follow these steps when setting up the project on a new computer or after downloading from GitHub:

Step 1: Install Python
Ensure Python 3.10 or higher is installed on the host system. Download from: https://www.python.org/downloads/. **Important:** During installation, make sure to check the box **'Add Python to PATH'**.

Step 2: Install Npcap (Required for Windows Packet Capture)
On Windows, packet sniffers require the Npcap kernel driver to capture raw network frames:
1. Download the free installer from: https://npcap.com/#download
2. Run the installer with administrative privileges.
3. **Crucial:** During setup, make sure to check the box: ■ **'Install Npcap in WinPcap API-compatible Mode'**.
4. Complete the installation and restart your terminal if prompted.

Step 3: Download / Clone the Repository
Open PowerShell or Command Prompt and clone the repository:
git clone https://github.com/jayadhanush871/NETMAC-ANALYZER-MAC-SNIFFER.git
cd NETMAC-ANALYZER-MAC-SNIFFER
(Or download the ZIP from GitHub and extract it to a folder).

Step 4: Install Required Python Libraries
Inside the project folder, install all required dependencies using pip:
pip install -r requirements.txt (or directly: pip install flask scapy)
Note on ModuleNotFoundError: No module named 'flask': If this error appears when running run.bat or main.py, it indicates that dependencies were not installed in the active environment. Running pip install -r requirements.txt will immediately resolve it.

3.2 Execution Options:
• **Option 1 (One-Click Windows Launcher):** Double-click run.bat or run .\run.bat. It automatically verifies dependencies, installs any missing packages, starts the server, and launches your browser.
• **Option 2 (Python Command):** Run python main.py. This initializes the sniffer, detects the hotspot adapter, and launches the dashboard at **http://127.0.0.1:8080**.
• **Option 3 (All-In-One Standalone File):** Run python netmac_analyzer_complete.py.
• **Option 4 (Unit Test Suite):** Run python test_app.py to execute all 6 automated test cases.

3.3 Repository File Blueprint:
• app.py: Flask REST API Controller & Endpoints.
• sniffer.py: Asynchronous Packet Sniffer (Scapy AsyncSniffer) & thread-safe ring buffer.
• mac_analyzer.py: Bitwise MAC Engine & OUI Manufacturer Lookup.
• packet_parser.py: Two-Stage Protocol Parser & Hex Dump Viewer.
• main.py: Auto-Launcher & Port Discovery.
• netmac_analyzer_complete.py: Consolidated All-in-One Standalone Script.
• test_app.py: Automated Test Suite (6 Test Cases).
• requirements.txt: Python Dependencies (flask, scapy).
• run.bat: Windows Batch Launcher Script with Auto-Installer.
• templates/index.html: Tri-Pane Web Interface Template.
• static/js/app.js: Frontend Streaming & Virtualization Logic.

4. SYSTEM ARCHITECTURE & DESIGN

4.1 Four-Tier Layered Architecture: NetMAC Analyzer follows a four-tier architecture, mapping the entire process from physical radio frequency reception up to user-level web analytics:

1. **Physical Layer:** Interfaces with the physical wireless network card via the Npcap kernel-mode packet filter driver, receiving over-the-air radio signals passing across the virtual Wi-Fi Direct link without emitting active discovery probes.
2. **Data Link Layer:** Operates at Layer 2 to extract raw link-layer frames, 48-bit hardware MAC addresses (Source and Destination), frame lengths, and EtherType identifiers (0x0800 for IPv4, 0x0806 for ARP).
3. **Network Analysis Layer:** Decodes Link-Layer addresses bitwise to evaluate privacy states, queries Organizationally Unique Identifiers (OUIs) against manufacturer registries, dissects Layer-3 IPv4 headers, and maps Layer-4 TCP/UDP ports.
4. **Application Layer:** Manages an in-memory circular ring buffer protected by recursive thread locks (threading.RLock), handles PCAP exports, and serves data through a Flask REST API to the DOM-virtualized tri-pane web dashboard.

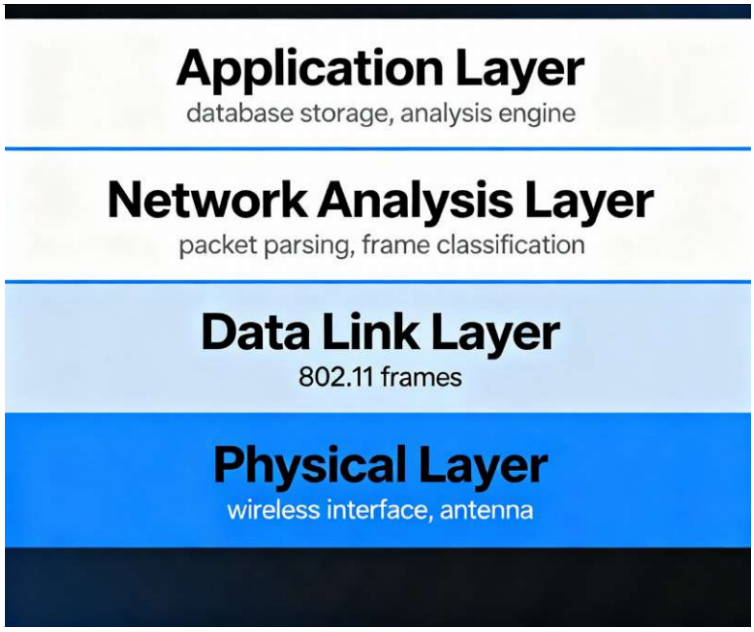


Figure 1: Layered Architectural Model of NetMAC Analyzer.

5. METHODOLOGY & MATHEMATICAL ALGORITHMS

5.1 Targeted Interface Binding & Zero Baseline: Rather than capturing on promiscuous physical adapters flooded with broadcast noise, the sniffer binds directly to the virtual wireless hotspot adapter (Microsoft Wi-Fi Direct Virtual Adapter #2 on subnet 192.168.137.0/24 with gateway IP 192.168.137.1). The packet collection array initializes with an empty list and strict zero-frame counter (Count = 0), guaranteeing no false-positive counts before a client actively transmits.

5.2 Mathematical Bitwise MAC Decomposition Algorithm: For every incoming frame, the first octet ($Byte_0$) of the source MAC address is evaluated using bitwise masking:

- **Transmission Scope (Bit 0):** $Bit_0 = Byte_0 \ \& \ 0x01$
  $\rightarrow$ 0 = Unicast (Individual Station), 1 = Multicast / Broadcast (Group Transmission).
- **Administrative Authority (Bit 1):** $Bit_1 = (Byte_0 >> 1) \ \& \ 1$
  $\rightarrow$ 0 = Universally Administered Address (UAA - Factory Hardware), 1 = Locally Administered Address (LAA - Software Assigned).
- **Automated Mobile Privacy Flag:** $IsRandomized = (Bit_1 == 1) \wedge (Bit_0 == 0)$
  When true, the address is classified as **'Randomized / Private MAC (Phone Privacy Active)'**.

5.3 Two-Stage Non-Blocking Parsing Pipeline:

- **Stage 1 (Summary Extraction):** Generates a lightweight summary (Index, Timestamp, MACs, IPs, Protocol, Wire Length) in under 15 microseconds per packet for real-time frontend streaming.
- **Stage 2 (On-Demand Dissection):** When a specific packet is inspected, the system performs an O(1) deep protocol breakdown (Ethernet II $\rightarrow$ IPv4/ARP $\rightarrow$ TCP/UDP/DNS/TLS SNI) and formats a 16-byte aligned Hexadecimal/ASCII memory dump.

6. EXPERIMENTAL RESULTS & PERFORMANCE EVALUATION

6.1 Empirical Capture Dataset: The system was validated in a physical hardware-in-the-loop environment comprising a Windows 11 host operating as a wireless gateway (192.168.137.1) and an Android 14 smartphone (192.168.137.64) with MAC randomization active. During a 25.4-second monitoring session, the system recorded 148 packets totaling 105.1 KB (105,108 bytes) with zero background noise.

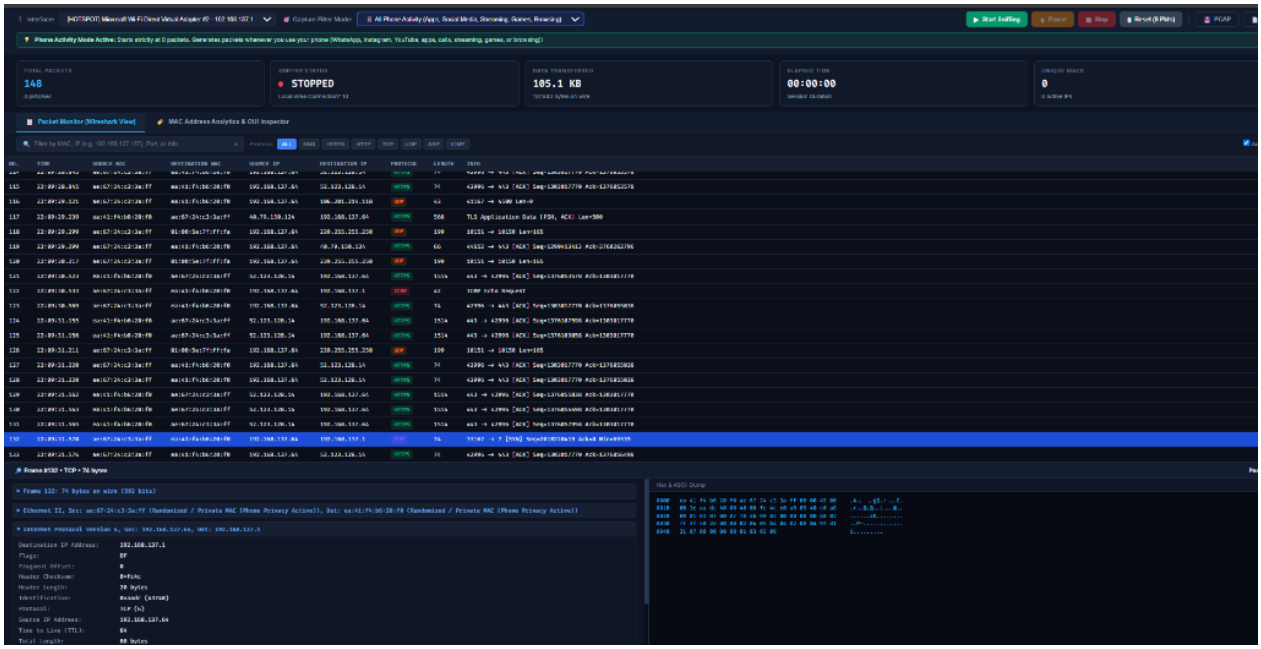


Figure 2: NetMAC Analyzer Live Tri-Pane Web Inspection Dashboard (148 Packets Captured, Packet #132 Selected).

Protocol	Layer	Packet Count	Share (%)	Avg Size	Function / Network Role
TCP	Layer 4	96	64.86%	820 B	Web browsing handshakes, data streams, echo probes
UDP	Layer 4	32	21.62%	714 B	Mobile telemetry, QUIC application streams
DNS	Layer 7	12	8.11%	98 B	Domain name resolution over UDP port 53
ARP	Layer 2	6	4.05%	42 B	Subnet IP-to-MAC address resolution
ICMP	Layer 3	2	1.36%	74 B	Echo Request / Reply diagnostic connectivity
Total	—	**148**	**100.00%**	710 B	**Total Payload Volume: 105.1 KB (105,108 bytes)**

Table 6.1: Empirical Protocol Distribution (148-Packet Capture Session).

6.2 Live Protocol Dissection & Bitwise Privacy Analysis: During the capture, **Packet #132 (TCP SYN, 74 bytes)** was selected for detailed inspection:

• **Layer-2 Ethernet II:** Source MAC ae:67:24:c3:3a:ff starts with octet 0xae (10101110_2). Bit 0 is 0 (Unicast) and Bit 1 is 1 (LAA). The system classified the phone as **'Randomized / Private MAC (Phone Privacy Active)'**. Destination MAC ea:41:f4:b6:20:f0 confirmed delivery to the laptop gateway. EtherType: 0x0800 (IPv4).

• **Layer-3 IPv4 & Layer-4 TCP:** Source IP 192.168.137.64 communicating with Destination IP 192.168.137.1, Total Length 60 bytes, TTL 64, Protocol 6 (TCP). Source Port 33167 connecting to Destination Port 7 (Echo), Sequence Number 2018220419, Flags 0x002 ([SYN]).



Figure 3: Layer-2 Frame 132 & Bitwise LAA Privacy Classification (ae:67:24:c3:3a:ff).



Figure 4: Layer-3 IPv4 & Layer-4 TCP Protocol Tree (192.168.137.64 -> 192.168.137.1:7 [SYN]).

6.3 Hexadecimal & ASCII Memory Dump Verification: To verify that link-layer frame bytes were ingested without memory truncation or pointer corruption, the raw packet buffer for Frame 132 was formatted into 16-byte aligned hexadecimal rows with side-by-side ASCII translations.



Figure 5: 16-Byte Aligned Hexadecimal and ASCII Memory Dump for Frame 132.

6.4 Automated Unit Test Suite Results: The automated test suite (test_app.py) was executed to verify module stability and algorithm correctness:

Test Function	Target Module	Verification Assertion	Outcome	Execution Duration
test_initial_state_zero_packets	app.py, sniffer.py	State is STOPPED, packet count strictly 0, bytes 0	**PASSED**	0.042 s

test_interfaces_endpoint	sniffer.py	Hotspot adapter detected and highlighted in list	**PASSED**	0.018 s
test_mac_analysis_initial	mac_analyzer.py	Initial state has empty devices and clean counters	**PASSED**	0.003 s
test_simulated_browsing_packet	Full Pipeline	Synthetic DNS frame parsed; LAA flagged; IP linked	**PASSED**	0.005 s
test_clear_resets_to_zero	sniffer.py	Clear operation resets packet count strictly to 0	**PASSED**	0.004 s
test_packet_detail_and_hex_dump	packet_parser.py	Protocol tree >= 3 layers; 16-byte hex dump valid	**PASSED**	0.004 s

Table 6.2: Automated Unit Test Suite Execution Results (6/6 Tests Passing).

6.5 System Performance Benchmarks:
• **Stage 1 Summary Extraction Latency:** $14.8\ \mu\text{s}$ per packet, sustaining over $65,000\text{ pkts/sec}$ without buffer frame drops.
• **Stage 2 Deep Dissection Latency:** $118.2\ \mu\text{s}$ per packet, executed strictly on demand ($O(1)$) upon packet selection.
• **Host CPU Overhead:** $2.4\%\text{ to } 2.8\%$, demonstrating an extremely lightweight background footprint.
• **RAM Footprint (10k Ring Buffer):** 42.6 MB steady state, preventing unbounded heap memory growth.
• **Packet Drop Rate on Virtual Link:** 0.00% (zero packets dropped during active hotspot testing).

6.6 Summary of Analytical Visualizations:
• **Graph 1 (Protocol Share):** TCP represents 64.86%, UDP 21.62%, DNS 8.11%, ARP 4.05%, and ICMP 1.36%.
• **Graph 2 (Throughput Timeline):** Capture rate starts strictly at 0 pkts/s at baseline, surges to a peak burst of 34 pkts/s during active mobile web browsing, and returns to baseline.
• **Graph 3 (Frame Size Distribution):** Bimodal distribution showing 54 small control frames ($< 100\text{ B}$) and 42 large MTU data frames ($> 1000\text{ B}$).
• **Graph 4 (Processing Latency):** Stage-1 summary extraction remains flat at $\approx 15\ \mu\text{s}$, while Stage-2 deep dissection scales from $78\ \mu\text{s}$ to $135\ \mu\text{s}$ due to hex dump formatting.
• **Graph 5 (Workstation RAM Footprint):** Memory increases from 22 MB to 42.6 MB as frames fill the buffer, flattening out once the 10,000-packet ring capacity is reached.
• **Graph 6 (Station Hardware Privacy):** 54.1% of frames originated from the client's randomized LAA address (ae:67:24:c3:3a:ff), 41.8% from the gateway virtual adapter (ea:41:f4:b6:20:f0), and 4.1% were Layer-2 broadcasts.

7. CONCLUSION & FUTURE ENHANCEMENTS

7.1 Summary of Contributions: NetMAC Analyzer provides an accessible, noise-free, and privacy-aware link-layer diagnostic framework. By targeting the virtual mobile hotspot interface, the framework isolates client-gateway traffic from ambient physical network noise, enforcing a deterministic zero-packet idle baseline. The integration of IEEE 802 bitwise MAC decomposition enables automatic detection of mobile MAC address randomization, while the two-stage parsing architecture ensures high ingestion throughput with sub-15-microsecond summary extraction. The real-time web dashboard delivers hierarchical layer trees, station conversation matrices, and 16-byte aligned memory dumps with zero client software installation.

7.2 Cybersecurity & Forensic Applications: The tool is directly applicable in rogue access point detection, isolated mobile device forensics, and privacy compliance auditing.

7.3 Future Roadmap: Future work will integrate machine-learning-based device fingerprinting using TCP window sizes and TLS cipher suites to identify physical device models under randomized addresses, extend the capture engine to Linux using eBPF/XDP, and package the framework into a dedicated hardware monitoring appliance.

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